

voltage DC power supply producing an output voltage in the range of 800V to 1500V depending upon the characteristics of the flash lamp and the required magnitude of simmer current. A high-voltage trigger pulse, typically 10kV to 15kV, creates pre-ionisation before the simmer power supply can take over and deliver the keep-alive current through the flash lamp.

The output of the simmer power supply is applied to the flash lamp through a series resistor called ballast resistor. The magnitude of the simmer current therefore depends upon the difference between the simmer supply output voltage and voltage across the flash lamp in the simmer mode, and the value of the ballast resistance. The value of the ballast resistance should be slightly higher than the negative impedance offered by the flash lamp in the simmer regime.

Simmer power supply is generally designed around an externally-driven flyback converter topology with output power delivery capability equal to product of required open circuit output voltage and magnitude desired simmer current. Open circuit voltage is further equal to sum of voltage across the lamp in simmering condition and voltage drop across the ballast resistor. Fig. 4 shows simmer power supply interface with the flash lamp.

A TTL/CMOS pulse applied to the gate of SCR switches it on, thereby producing high-voltage trigger pulse across the secondary winding of trigger transformer. High-voltage trigger produces required pre-ionisation, forcing simmer current to flow through it.

This circuit, however, has a drawback that if simmer current stops due to some reason, there is no in-built mechanism to restore it. This shortcoming is overcome in the circuit schematic of Fig. 5. In this case, if the simmer current stops, an astable multivibrator controlled by a comparator restores normal operation. The astable multivibrator operates typically at 20Hz to 30Hz. Simmer power supply modules like many other laser electronics subsystems, such as capacitor-charging power supplies

and flash lamp trigger circuits, are also commercially available for OEM manufacturers. One such module is shown in Fig. 6.

Pulse-forming network

The pulse-forming network (PFN) produces a critically damped current pulse through the flash lamp when the energy storage capacitor is discharged through it. This is the most efficient



Fig. 6: Simmer power supply module

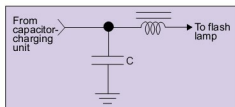


Fig. 7: Single-mesh pulse-forming network (PFN)

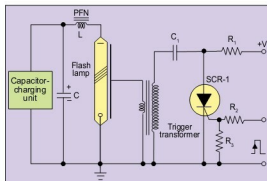


Fig. 8: External triggering circuit

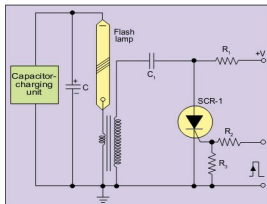


Fig. 9: Series triggering circuit

way of energy transfer, which also minimises reverse voltage appearing across the capacitor.

Fig. 7 shows single-stage PFN. The value of energy storage capacitor depends upon stored energy (E_0), desired pulse width (t_p) and flash lamp impedance parameter (K_0), and is given by $C = (0.09 E_0 t_p^2 K_0^{-4})^{1/3}$. Also, $t_p = \sqrt{LC}$ and $L = t_p^2 / 9C$.

Flash lamp trigger circuit

Flash lamp trigger circuit is required only in case of non-simmer mode of operation. Common modes of flash lamp triggering include overvoltage triggering, external triggering, series triggering and parallel triggering. Overvoltage and parallel triggering schemes are less popular. External and series triggering circuits are more common and are shown in Figs. 8 and 9, respectively.

The main advantage of external triggering is that it does not interfere with the main energy discharge circuit. The disadvantage is that high-voltage trigger point is exposed and therefore needs to be properly isolated from the environment, lest it causes problems in high altitude or humid conditions.

External triggering is recommended for low repetition rate, low energy systems where the flash lamp is air cooled.

In the case of series triggering circuit, the series trigger transformer is designed in a way that the transformer core saturates and the saturated secondary winding inductance serves the purpose of the PFN inductor also. Series triggering offers the advantages of reliable and reproducible triggering.

Receiver electronics of laser rangefinder

Fig. 10 shows the block schematic arrangement of different building blocks of receiver electronics of a typical laser rangefinder. Optoelectronics front-end circuit is one of the most critical building blocks of the receiver section. It is supposed to transform the received laser pulse, which could be anywhere in the